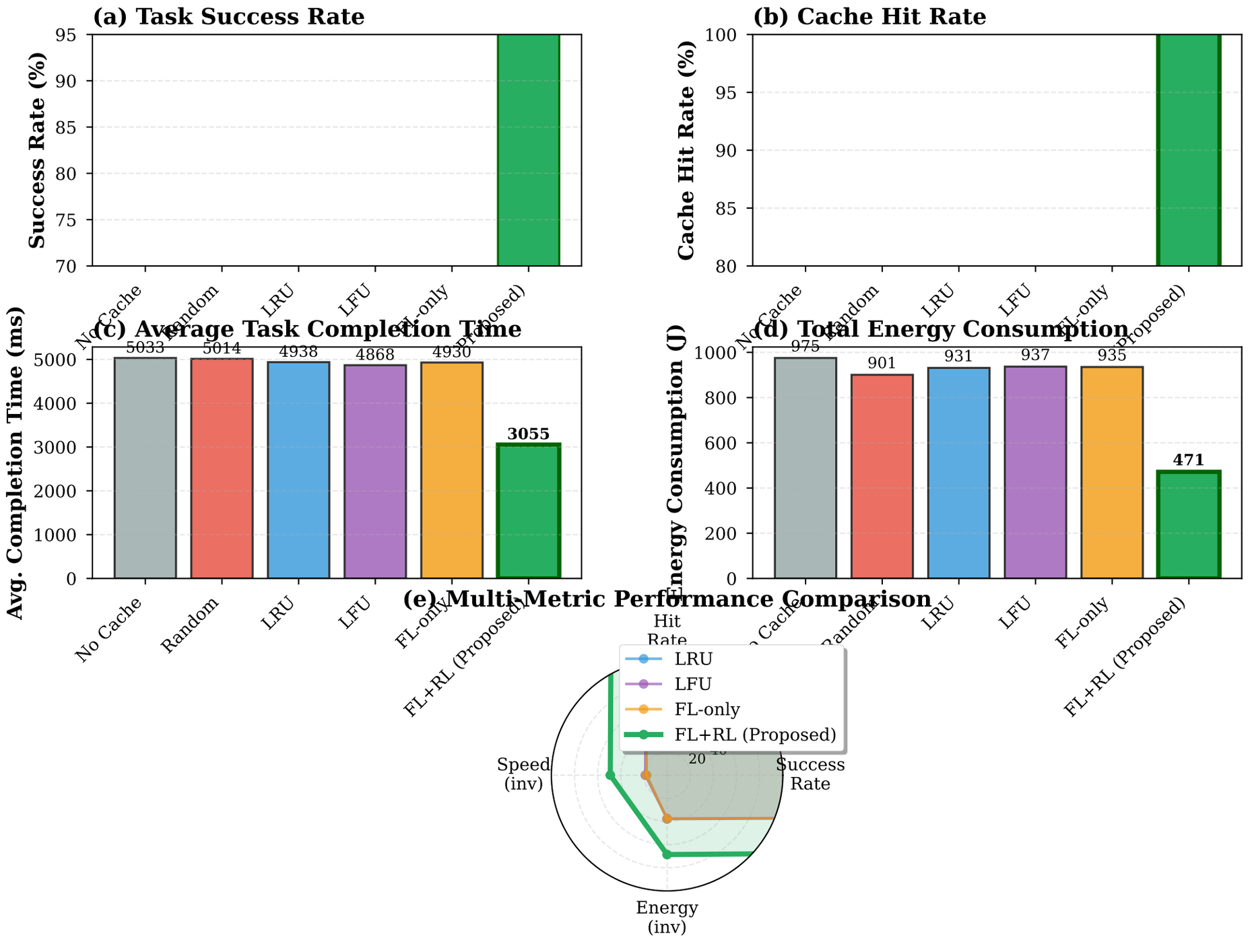
FL+RL Performance Comparison for VANET Caching
(a) Task Success Rate
Success Rate (%)
95
90
85
80
75
70
(b) Cache Hit Rate
Cache Hit Rate (%)
100
95
90
85
80
(c) Average Task Completion Time
Avg. Completion Time (ms)
5033
5014
4938
4868
4930
3055
5000
4000
3000
2000
1000
0
(d) Total Energy Consumption
Energy Consumption (J)
975
901
931
937
935
471
1000
800
600
400
200
0
No Cache
Random
LRU
LFU
FL-only
FL+RL (Proposed)
(e) Multi-Metric Performance Comparison
Hit Rate
Success Rate
Energy (inv)
Speed (inv)
20
LRU
LFU
FL-only
FL+RL (Proposed)